Supplementary Material for "UTC Timestamping of a Free-Running Delta–Sigma ADC With Evaluated Uncertainty: Method and Validation on the ozmV Synchrophasor Instrument"

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***Abstract*—This document details the measurements that the paper summarizes: the laboratory setup and its passes (S-I), the determination of the ADC delay and the checks of its fit (S-II), the two prototypes measuring one signal with independent receivers (S-III), the effect of the timestamping method on recorded data (S-IV), the numerical P-class tests, the reporting latency and the computational load (S-V), the holdover through GNSS outages (S-VI) and the start-up window (S-VII). The data behind every table and figure, and the programs that produce them, will be in the repository that accompanies the paper (https://github.com/paketecuen/ozmv-pmu-timing). Section numbers of the paper are cited as Section II, III, etc., and those of this document as S-I to S-VII.**

S-I. Laboratory Setup and Measurement Passes

Two prototypes of ozmV, A and B, each with a GNSS receiver (QUESCAN M10TTL, u-blox UBX-M10050) whose antenna is integrated in the module housing, measured the same phase of the laboratory mains through the phase-A voltage channel. The PPS of the receiver was applied to the PRU of the prototype and, through a resistive divider (5.6 kΩ in series, 1 kΩ to ground), to the neutral-current input of the ADC (connector CN5, input AIN0 of the ADS131M08), which the waveform stream recorded in ADC volts at 31 502 samples/s together with the per-sample time track. The measurements were made on two days (2026-10-27 and 2026-10-29) on the 230 V mains of the laboratory of the Department of Engineering of the University of Almería, and Table S-I lists the passes.

A. The Reference Digitizers

A Fluke 8588A digitized the mains voltage (range 1000 V) and a Fluke 8558A, triggered by the 8588A through the trigger cable, digitized the PPS (range 10 V). Both operated in digitize mode with a sampling interval of 50 µs, an aperture of 49 µs and the input filter off, in bursts of 200 000 readings (10 s), 60 bursts per M2 pass and 6 per M3 pass. The host computer kept time by NTP, which only identifies the UTC second of the PPS edges. For M3, the same PPS was applied to both instruments through short leads with series resistors.

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TABLE S-I
Passes of the Two Laboratory Sessions (UTC)

Day	UTC	Pass	8558A PPS
1	08:23–08:43	M3, slave lag, 10 V and 1000 V, twice	B
1	08:55–09:36	M2-A, voltage delay of A, twice	A
1	11:02–11:06	M3, trigger output on the aperture start	A
1	11:14–11:33	M2-A, same	A
1	11:38–11:57	M2-B, voltage delay of B	B
1	12:06–12:28	M3, both ranges, after M2	B
1	07:50–12:49	M1 on both, and A against B, throughout	
2	08:02–10:21	M1 on both; branches swapped at 09:49	
2	09:51–10:21	M4-r02: A against B, nothing else on the PPS lines	
2	10:53–11:33	E4-A: A loses its signal for 1, 5 and 10 min	
2	11:38–12:42	E4-B: B loses its signal for 1, 5 and 10 min	

The trigger output of the 8588A has two modes. In the default mode, a pulse at the end of the acquisition (`ROUT:TOUT ACO`) starts the 8558A one sample interval after the 8588A. The slave then lags by 49.3 µs, one aperture plus the residual lag, and an analysis that pairs edges within half a sample interval pairs none. With the output configured as a square wave active during the acquisition (`APE`, falling edge to the external trigger of the slave), both instruments acquire together. Table S-II reports all the M3 passes. The residual lag is 112 to 124 ns with the master on 10 V and 315 to 326 ns on 1000 V, the 1000 V range of the master adding about 200 ns. The M2 analysis uses 325.8 ns with APE (the weighted mean on 1000 V) and 49 327.0 ns with ACO. The per-edge dispersion, 100 to 310 ns RMS, exceeds the 82 ns of two independent 200 ns cells, and is larger on the 1000 V range, where the 2.3 V pulse is a small fraction of the range.

B. The Measurement of the Channel Delay (M2)

A reading of the aperture is the mean of 246 samples taken 200 ns apart, so the reading that straddles a PPS edge places the edge within one 200 ns cell. Since the time base of the digitizers is not locked to the PPS, the edges of successive seconds fall at different phases of those cells. The mean

TABLE S-II
M3: LAG OF THE 8558A BEHIND THE 8588A, BOTH DIGITIZING THE SAME PPS

Pass	Range	Trigger	Edges	Lag (ns)	u (ns)
m3-10V-antes	10 V	ACO	53	49 124.5	21.0
m3-10V-antes-r02	10 V	ACO	59	49 183.1	14.7
m3-1000V-antes	1000 V	ACO	57	49 329.8	32.2
m3-1000V-antes-r02	1000 V	ACO	54	49 322.2	42.4
m3-10V-ape	10 V	APE	58	124.1	12.9
m3-1000V-ape	1000 V	APE	52	315.4	38.2
m3-10V-despues-ape	10 V	APE	57	112.3	13.3
m3-1000V-despues-ape	1000 V	APE	552	326.4	8.9

TABLE S-III
M2: DELAY OF THE PHASE-A VOLTAGE TIMESTAMPS OF THE PROTOTYPE BEHIND THE PPS-REFERENCED TIME BASE OF THE 8588A

Pass	Prot.	Trigger	Windows	Delay (µs)	RMS (ns)	u (ns)
m2-A-VA	A	ACO	2895	72.159	47	11.2
m2-A-VA-r02	A	ACO	2768	72.174	47	12.9
m2-A-VA-ape	A	APE	2940	72.225	47	4.9
m2-B-VA-ape	B	APE	2882	72.713	94	8.0

of many edges is therefore unbiased, and its standard error decreases as $1/\sqrt{\text{edges}}$ from the 58 ns RMS of one cell. The ten edges of a burst fix the time base of that burst to about 20 ns, and an edge that falls in the 1 µs gap between apertures is dropped. The phase of the 50 Hz voltage with respect to UTC is obtained from the readings of the 8588A on that time base (minus the lag of the slave, plus half the aperture) and from the samples of the prototype with their timestamps. The two phases are compared in windows of 0.2 s by a four-parameter sine fit, which gives the delay by which the timestamps of the prototype lag the input. Table S-III reports the four passes. The delay measured in this way is the phase delay of the channel at 50 Hz. At other frequencies it differs wherever the phase response of the front end is not linear, and the waveform stream, which also needs it, is outside this measurement. A Rohde & Schwarz RTE1000 oscilloscope with both probes grounded at the PPS node showed no difference between the edge at the PPS wire of the prototype and at the input of the 8558A at 100–200 ns per division. The check at the pin of the processor was not possible, since the pin is hidden under the module.

C. Loads on the PPS Line

The PPS of the receiver was shared, at times, with the 8558A and with the probes of the oscilloscope. Whenever an instrument was connected to one of the two PPS lines, the phase difference between the phasors of the two prototypes (S-III) moved by 100 to 300 ns. It stepped by −0.27 µs at 11:35 of the first day, when the 8558A moved from the pulse of A to that of B, and by about 0.3 µs at 08:36 of the second day, when the probes of the oscilloscope were connected to the pulse of A. A load on the line delays the edge seen by the capture input of the prototype. The reference digitizer on the same node sees the same delayed edge, so the delays of Table S-III are unaffected, whereas the time with respect to UTC is.

TABLE S-IV
M1: OFFSET OF THE SAMPLE TIMESTAMPS WITH RESPECT TO THE INPUT, BEYOND THE GROUP DELAY OF 190.5 t_{MOD}

Prot.	Day	Recordings	Edges	Offset (ns)	Spread (ns)
A	1	10	16 493	11 876.9	20.2
B	1	9	15 240	11 899.5	28.6
A	2	4	7103	11 867.9	30.6
B	2	4	7137	11 830.0	33.5

Hence, the specification of the receiver holds for a pulse that drives the input of the prototype alone. The acquisition faults of the prototype boards while cables were being changed, and the reboots they caused, are matters of the prototypes and do not enter the measurements.

S-II. THE ADC DELAY

A. Simulation of the Measurement

It may seem surprising that a 31.5 kS/s ADC can locate an edge to nanoseconds. Each output sample, however, is a weighted average of 382 modulator samples taken every $t_{\mathrm{MOD}} = 248$ ns, so the amplitudes of the samples around one edge locate it to within about 72 ns RMS, the quantization of the modulator grid, and the edges of successive seconds fall at different phases of that grid. The measurement was simulated (`analysis/m1_pps_step_sim.py`) with the modulator and the sinc^3 filter of the ADC, the noise of its data sheet (21.3 µV RMS at this gain and rate) and the measured errors of the sample timestamps (the prediction error of the counter-to-UTC mapping and the residual of the envelope regression). The simulation recovers an offset of $44\,t_{\mathrm{MOD}}$ with an RMS error of 2 ns from 600 edges with a sharp edge at the input, and of 6 to 13 ns when the RC network has to be fitted as well.

B. The Measured Offset, Recording by Recording

Every 30-minute recording of both days with at least 1000 usable edges and the pulse of the prototype itself on its input was fitted separately (`analysis/s01_wmu_batch.py`, which runs `analysis/m1_pps_delay.py` on each). Table S-IV reports the means per prototype and day, around which the recordings of one day are dispersed by 20 to 34 ns. The pulse was 338 mV at the ADC input and the noise of the samples 0.2 mV. In one recording, the 1733 edges fell on 3930 distinct phases of the sample period, never more than 175 ns apart. The time constant of the one-pole input network is 32.8 (A) and 32.7 µs (B), compared with about 27 µs from the nominal series resistors and capacitor of the input. The difference is the source resistance of the divider that feeds the pulse, which the fit absorbs. The model-free centroid of the equivalent-time impulse response, delay plus time constant, is 44.73 (A) and 44.64 µs (B).

C. Checks of the Fit

On one recording (prototype B, first day, 08:48–09:18 UTC, 1733 edges, analyzed with `analysis/m1_models.py` and `analysis/m1_windows.py`):

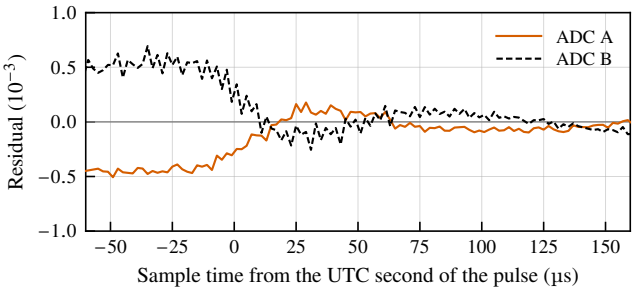


Fig. S1. Residual of the fits of Fig. 2 of the paper (recordings A-20261027T082028Z, 1746 edges, and B-20261027T091857Z, 1780 edges), for each ADC, in bins of 2 µs.

TABLE S-V
B MINUS A: THE TIME BY WHICH THE PHASOR OF B LAGS THAT OF A, PER PERIOD

Period	Seconds	Mean (ns)	RMS (ns)	5-min means (ns)
Day 1, instruments on the PPS lines	15 765	584.8	209	280–1060
Day 2, 08:12–09:14, same	3477	746.7	182	453–960
Day 2, M4-r02, nothing on the lines	1756	504.8	72	473–529
Day 2, E4, cross PPS cabling	5977	499.7	115	361–765

- the level windows of the fit, which define the low and high states of the pulse, move the offset between 11 867 and 11 932 ns (the windows of the paper give 11 879), and the two halves of the recording give 11 896 and 11 862 ns;
- a second pole added to the input network collapses to 1 ns and leaves the offset at 11 878 ns;
- a direct feed-through added to the one-pole network comes out at $-0.9\,\%$ of the step, unphysical for a passive network, with the offset at 11 594 ns; the one-pole model is kept.

On this recording, the residual of the fit is 6.4×10^{-4} of the step RMS. Fig. S1 shows the residual of the fits of Fig. 2 of the paper, i.e., of the recordings with 1746 and 1780 edges, for both ADCs in bins of 2 µs. On the basis of these checks, $u_D = 40\,\text{ns}$ is adopted in the paper.

D. Check of the Time Base With Respect to the Pulse, Second by Second

The same edges, fitted one second at a time with the time constant fixed at the value of the recording, give the time of each edge from the timestamps of the samples around it. Over the recording above, that time is dispersed by 46 ns RMS from one second to the next about its median. On the bare sample grid (sample index times a constant period, which removes the timestamps and leaves the pulse and the measurement) the dispersion is 33 ns, and the difference between the two, the second-to-second noise of the timestamps themselves, is 15 ns (`analysis/s01_grid_vs_labels.py`). The counter-to-UTC mapping of Section II-C was chosen on the same basis. On the test bench, the two-point line predicted the next pulse with an RMS error of 14.6 ns, least-squares lines through 3, 4 and 8 reference points performed similarly, 12.2 to 14.9 ns, and a line through 16 reference points performed much worse, 35.1 ns. The reason is that over more than a few seconds the wander of the oscillator frequency matters more than the jitter of the pulses, about $\pm 10\,\text{ns}$ from one pulse to the next.

E. Timing of the NMEA Sentences

With the receiver of the test bench, the RMC and GGA sentences followed their pulse by 128–133 ms, well within the 900 ms association window of Section II-C. This sentence-after-pulse timing is the property of the receiver on which the validation of the UTC-second assignment relies, and it has to be confirmed for the receiver model in use.

S-III. TWO PROTOTYPES ON ONE SIGNAL

The phasors of the two prototypes were paired by their UTC timestamps, never by their arrival, and their waveform streams were compared on the same UTC seconds through their time tracks (`analysis/s01_ab_pmu.py`, `analysis/s01_ab_wmu.py`). Table S-V reports the difference, i.e., the phase of B behind that of A expressed as a time, in each period. For the same seconds, the two streams of one prototype give the same difference to 2.0 ns RMS (mean 0.0 ns). The amplitude ratio B/A was 0.999 69, the frequency difference 1.0×10^{-4} Hz RMS and the ROCOF difference 0.023 Hz/s RMS. In the clean period, with nothing but the two prototypes on the PPS lines and the voltage branches swapped between them (the difference did not move with the swap, so it lies in the channels of the prototypes and not in the wiring), B lags A by 504.8 ns. The channel delays of Table S-III differ by 487.4 ns, and the residual, 17 ns, is the disagreement of the two receivers and time chains, with the caveat that two receivers of one model may share a bias with respect to UTC. During 4.4 h of the first day, with the reference digitizers connected to one PPS line or the other, the difference wandered between 0.28 and 1.06 µs (Table S-V).

S-IV. EFFECT OF THE TIMESTAMPING METHOD ON RECORDED DATA

To isolate the effect of the timing, 40 s of mains voltage recorded by the prototype together with the per-sample timestamps (release 17 of the software, 2026-10-14) were timestamped again, in simulation, with the methods reviewed in Section I of the paper. Each version was run through the same estimator (`analysis/e3_dating_strategies.cpp` and Fig. 4 of the paper). The errors are differences from the reports computed with the per-sample timestamps, and therefore measure the effect of the timestamping alone.

Timestamping the first sample and extrapolating with the nominal period accumulates the 40 ppm of the crystal: 1.61 ms in 40 s, a TVE of 50 % and an RFE of 0.42 Hz/s. Re-alignment at each PPS limits the error to the drift of one second, 40.3 µs, which is still a TVE of 1.2 % just before each pulse and an

RFE of 0.023 Hz/s. Moreover, the jump at each pulse breaks the windows that straddle it, so that 120 of the 1997 reports were lost and the estimator restarted 40 times. With the period of the previous second, the drift almost disappears once a full second has been measured. From its third second on, this method stayed within 59 ns of the per-sample timestamp, less than the uncertainty of that timestamp, and its largest errors and its 6 lost reports fell in the first two seconds. The polled DRDY edges without the regression of Section II-B were simulated by adding to the per-sample timestamps a polling delay drawn uniformly from the measured range of 0 to 1.52 µs. They cost only 0.026 % of TVE but 0.079 Hz/s of RFE, about 20 % of the P-class limit, with no other error in the chain. Timestamps truncated to whole microseconds err by up to 1.00 µs and cost 0.016 % of TVE and 0.006 Hz/s of RFE.

S-V. Numerical P-Class Tests, Reporting Latency and Computational Load

A. The Two Parts of the Change

The change of the estimator described in Section IV-B of the paper had two parts: the phasor from the first-order sub-model instead of the full second-order model, and frequency and ROCOF from the positive sequence instead of phase A. A factorial run of the four combinations through the P-class tests (`analysis/e5_factorial.sh`, `data/e5/factorial`) separates their effects. The phasor order alone removes the step overshoot and sets the harmonic, modulation and ramp TVE of Table IV of the paper, and the reference alone sets the ramp RFE and the noise-limited RFE, each leaving the figures of the other untouched. The steps were evaluated as the standard prescribes: equivalent-time sampling over 20 step positions, state levels and $\pm 0.5\,\%$ boundaries for the response time, the delay time and the overshoot (5.2.3, 5.2.4 and Annex C.4 of IEC/IEEE 60255-118-1). Earlier figures of 40 ms of response time and 3.6 ms of delay, obtained before the harness followed the method of the standard, were artifacts of that harness.

B. Noise and Signal-to-Noise Ratio

With white noise 60 dB below the fundamental, the worst steady-state RFE is 0.30 Hz/s at 50 Hz and 0.41 Hz/s at 60 Hz from the positive sequence, compared with 0.51 and 0.66 Hz/s from phase A alone. The P-class limit of 0.4 Hz/s therefore requires an SNR of about 57 dB at 50 Hz and slightly above 60 dB at 60 Hz, a figure that the standard recommends PMUs fed by merging units to declare (Annex E.4.2 of IEC/IEEE 60255-118-1). The noise floor of the acquisition chain, 75 to 77 dB below the mains fundamental on unconnected voltage paths during a real recording, leaves a margin of more than 15 dB. The interharmonics of the mains itself are outside this figure.

C. Reporting Latency

The latency was measured on both prototypes during the M4-r02 pass of the second day, from the C37.118 frames that each prototype published to its own network interface and captured there with their arrival times (89 998 and 89 967 frames). The arrival time minus the report time of each frame is 31 to 52 ms from the 1st to the 99th percentile, with a median of 41.3 (B) and 41.1 ms (A) and maxima of 100 and 63 ms. The P class allows $2/F_s = 40\,\mathrm{ms}$. The half window of the fit is 20 ms and the half length of the decimation filter 4.1 ms. The rest comes from the batches of 256 samples (8.1 ms) in which the acquisition hands samples to the estimator, and from the scheduling of the estimation and the publishing. The arrival times are those of the system clock of the prototype, which the time service disciplines to the receiver, and this measurement does not verify that clock to better than a few milliseconds.

D. Computational Load

On one Cortex-A53 core at 1.4 GHz, the estimator takes 3.5 % of the core for three voltage channels at 50 frames/s, decimation filter included, about 0.7 ms per report with up to three carrier refinements. Fitting every sample instead of one in 16 would take 19 %. For comparison, a single-channel two-cycle Taylor–Fourier estimate at 8 kHz takes about 7 ms on a BeagleBone Black (Tosato *et al.*, 2018, cited in the paper). The whole measurement service of the prototype, which also computes the power-quality indices, takes 25.5 % of one of the four cores with the PMU off and 35.5 % with it publishing 50 frames/s. Of those ten points the estimator accounts for 3.5, and the rest goes to handing every sample, with its time and time quality, to the estimator and to publishing the reports. Streaming every sample with its time track adds another 14 points, so the prototype runs both at about half of one core.

S-VI. Holdover Through GNSS Outages

A. One Hour on the Test Bench

Fig. S2 summarizes one hour on the test bench, with the antenna under open sky, during which reception was lost three times, twice because the antenna was covered by hand. The PMU published 179 947 reports at 50 frames/s, one at every instant of the 20 ms grid, and its estimator never restarted. During the outages, which lasted 16.3, 36.8 and 10.7 s, the time-quality code of the reports followed the declared bound through the decades from 100 ns to 100 µs, and the unlocked-time bits changed after 10 s. The receiver came back 493, 1531 and 796 ns away from the holdover, no more than 15 % of the bound declared at each instant. The waveform stream flagged every sample of the three outages, with an uncertainty field that grew to 18.7 µs over the longest, and lost none. Before this design, a single outage had cost the PMU 2.96 s of reports and two restarts of its estimator. In an earlier run with the antenna in its deliberately poor position, 22 of 23 outages ended within 2.4 µs of the holdover, and the remaining one, after 8 min, ended 15.2 µs off, about 5 % of the bound at that length. Under open sky, over eight hours of logging, the receiver returned between 19 and 1324 ns from the holdover, with a median of 112 ns.

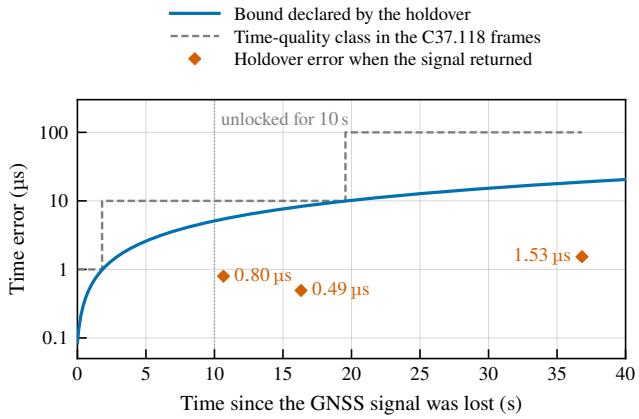


Fig. S2. Holdover through GNSS outages during one hour on the test bench. Solid line: the bound declared from the start of an outage with a fresh period. Dashed staircase: PMU time-quality class carried by the C37.118 reports during the longest outage. Diamonds: distance between the holdover and the GNSS time when the signal returned, for the three outages of the hour.

B. The Drift Allowance

The drift allowance of the bound, $\delta = 0.5\,\mathrm{ppb/s}$, comes from the test-bench crystal itself. Over ten hours of logging (`analysis/crystal_drift.py`), its frequency changed by up to 0.27 ppb/s averaged over an hour, 0.40 ppb/s over half an hour and, once, during a thermal transient, 0.68 ppb/s over ten minutes. Even that rate, sustained for a whole hour from an exact period, would leave the holdover 4.4 ms off, within the 5.0 ms that the bound declares by then. A return within 10 µs, a third of the sample period, continues the same timescale, whereas a larger one starts a new timescale and the waveform record marks the discontinuity as a gap.

C. The Six Outages of 1, 5 and 10 Minutes

Table S-VI reports, for each outage of Fig. 5 of the paper, its length (from the first to the last sample timed by the holdover), the error at its end and the largest error during it with respect to the pulse of the other prototype, and the bound declared at the end. It also reports the return check of the prototype itself and the error that a linear frequency drift would have accumulated, computed from the change of the measured period across the outage (`analysis/s02_e4.py`). The error is the offset of the timestamps with respect to the reference pulse minus the median offset over the five minutes of validated GNSS time before the outage, which removes the ADC delay and the constant difference between the receivers. The return check of the prototype is the distance between the holdover and the first validated sample at the return. Returns beyond 10 µs start a new timescale, which the waveform track flags, so these are the time steps of the prototype rather than the rejoins it counts within the same timescale. The antenna was covered by carrying it indoors into a closed box, which took 20 to 30 s. In one case (A, 10 min) the receiver saw the sky for 4 s during that transit without validating its time, and the outage is counted from the actual loss of the fix.

The shorter spontaneous holdovers of the two days, 45 of 10 to 29 s on the first day with the pulse of the prototype itself on its input, gave errors of at most 0.99 µs, never more than half the declared bound apart from one disturbed second while cables were being changed. For these holdovers, the return check of the prototype agreed with the independent measurement to 10 to 70 ns. One natural outage of 5.2 min on the first day, during which the receiver stopped its pulse, ended with the return check of the prototype at 38 ns with respect to a bound of 182 µs.

TABLE S-VI
E4: Error of the Holdover at the End of Each Outage, With Respect to the Pulse of the Other Prototype

Outage	Length (s)	Error at end (µs)	Largest (µs)	Bound at end (µs)	Own check (µs)	Drift pred. (µs)
A, 1 min	100	1.74	2.04	52.9	1.85	2.1
A, 5 min	349	13.75	13.75	206.1	13.63	10.6
A, 10 min	647	40.39	40.40	430.8	40.52	28.9
B, 1 min	98	3.57	3.57	51.6	3.42	1.4
B, 5 min	331	13.79	14.08	193.8	14.13	6.5
B, 10 min	660	22.05	22.10	440.6	21.97	7.9

S-VII. Measurements Before GNSS Time Acquisition

The test-bench board has no battery-backed clock. At each boot it restores the time saved at shutdown, and the time service steps the system clock by the length of the outage, about 14 s on the test bench, once it selects the receiver, some 25 s after the kernel starts. The measurement service starts earlier, about 15 s after the kernel. Labeled with the restored clock, the first records would be timestamped 14 s early, and before the change described here they were. In each of 20 reboots, one or two of the power-quality records, made every 3 s, were stored inside the real outage, and the stored data showed an outage 14 s shorter than it was.

The prototype now measures from its first sample and assigns the time afterwards. A session that starts before its time is known opens a provisional window, in which the samples are labeled by counting nominal periods from the first one and everything the analyzer would store is withheld, and the first sample with a validated UTC-second assignment closes the window. Because every conversion is counted, the final time of each withheld sample follows from the first validated sample by counting conversions back at the period measured with GNSS, so no sample is misnumbered and the error is the drift of the oscillator over the window, a few nanoseconds. The withheld records are stored with it. In 20 reboots (Fig. S3) the window lasted 5.3–6.3 s and was always closed by GNSS time. It moved the withheld records by 14.04–14.38 s, at a rate that corrected the 40 ppm of the crystal, no record was stored on the restored clock, and the stored data showed the true outage, 26.2–27.7 s.

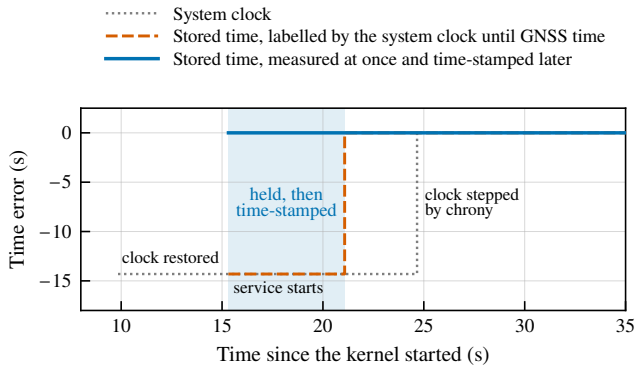


Fig. S3. Start-up after a reboot, a typical boot from the means of 20 (`data/boot-after-r20.csv`): the error of the time stored with each measurement, before the change (the restored clock) and after it (the provisional window, whose records are timestamped once the GNSS time is known).